Practical - 8

Practical Title: MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Problem Definition: Design the database for StudentHub with students, events and registrations. Create PHP database connection using MySQL statements, test PDO connectivity, and demonstrate prepared statements.

A. Theory & Analysis

1. Is the schema normalized properly?

Yes. The studenthub_db schema is structured into normalized tables: authentication credentials in 'users', student profiles in 'students', event records in 'events', student-event signups in 'registrations', and activity logs in 'audit_logs'. Splitting the data into distinct tables avoids redundant columns.

2. Are primary keys and foreign keys correctly defined?

Yes. Each table defines an auto-incrementing integer primary key (id). Relationships use foreign keys: students.user_id references users.id; events.created_by references users.id; and registrations references both students.id and events.id. A composite UNIQUE key (student_id, event_id) prevents duplicate registrations for the same event.

3. Is the database connection handled securely?

Yes. The connection script php/db_pdo.php creates a PDO instance with PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION, UTF-8 charset, and disabled emulate prepares. Parameterized prepared statements bind parameters safely, preventing SQL injection vulnerabilities.

B. Implementation Details

I designed the relational database schema in sql/schema.sql and imported it into MySQL. The schema contains 5 tables (users, students, events, registrations, audit_logs) seeded with 21 student records and 8 events. I created php/test_pdo.php to verify PDO database connectivity, query execution, and prepared statement parameter binding.

C. Evidence of Work

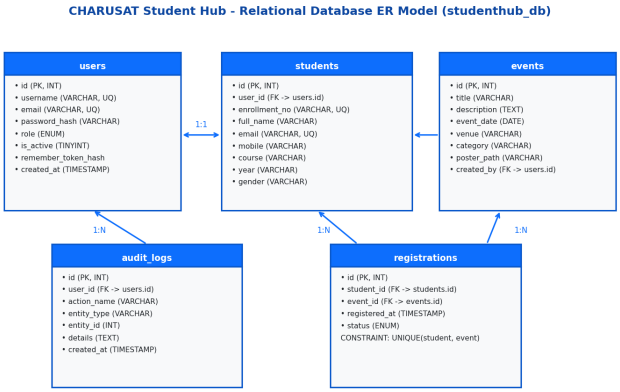


Figure 1: CHARUSAT Student Hub Relational Database ER Model Diagram

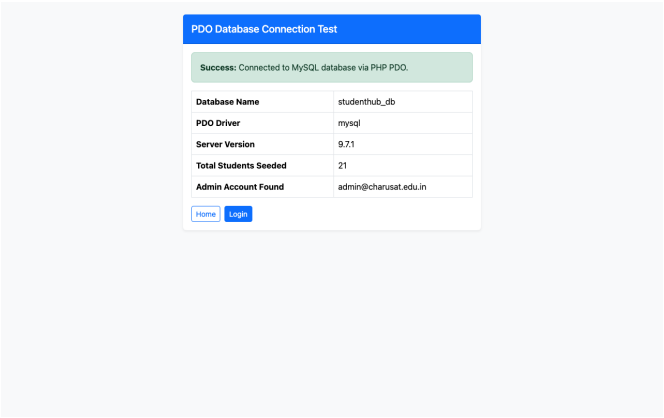


Figure 2: PHP PDO Database Connectivity Test and Table Diagnostic Output

D. Repository Link

GitHub Repository: https://github.com/neel92654/WDF_A_ARP

E. Learning Outcomes

I learned relational database design, foreign key constraint definition, and establishing secure database connectivity using PHP PDO with prepared statements.